

Semester End Examination Solutions

Database Management System (E1PA104B)

Section A: Fundamentals & Architecture

1. Define briefly DDL, DML, and DCL with examples.

- **DDL (Data Definition Language):** Defines the database structure. Example: `CREATE TABLE Student (ID int, Name varchar(50));`
- **DML (Data Manipulation Language):** Manages data within objects. Example: `INSERT INTO Student VALUES (1, 'Ravi');`
- **DCL (Data Control Language):** Controls access to data. Example: `GRANT SELECT ON Student TO User1;`

2. Comparison: File System vs. DBMS

Feature	File System	DBMS
Data Redundancy	High; multiple copies in different files.	Low; centralized data storage.
Data Consistency	Difficult to maintain across files.	Enforced via integrity constraints.
Data Sharing	Isolation makes sharing difficult.	Simultaneous access for multiple users.
Security	Basic OS-level file security.	Granular user-level access controls.
Data Independence	Low; programs depend on data structure.	High; physical/logical independence.

Section B: Normalization & Design

3. 1NF vs. 2NF

1NF: Requires atomic values and no repeating groups.

2NF: Must be in 1NF and ensure every non-prime attribute is fully functionally dependent on the primary key (no partial dependencies).

6. Normalization to 3NF

Relation: STUDENT (RollNo, StudentName, CourseID, CourseName, Instructor)

Transitive Dependency: RollNo → CourseID → CourseName, Instructor.

3NF Decomposition:

1. R1 (RollNo, StudentName, CourseID)
2. R2 (CourseID, CourseName, Instructor)

Section C: Transactions & SQL

7. Banking Transaction Analysis

- **Serializable:** No, the interleaved execution results in an inconsistent state.
- **Inconsistency Type:** Inconsistent Analysis (Reading intermediate/uncommitted results).
- **Preserving Order:** Executing T1 then T2, or T2 then T1.

8. SQL Queries

```
-- Students in Computer Science
SELECT Name FROM STUDENT WHERE DeptID = (SELECT DeptID FROM DEPARTMENT WHERE DeptName = 'Computer Science');

-- Students older than 20
SELECT RollNo, Name FROM STUDENT WHERE Age > 20;

-- Dept and HOD for RollNo 105
SELECT DeptName, HOD FROM DEPARTMENT D JOIN STUDENT S ON D.DeptID = S.DeptID WHERE S.RollNo = 105;
```